Qi Xu

Spatial World Models ▪ Embodied Intelligence ▪ Multimodal Learning

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EDUCATION

Westlake University 2026 – Present
PhD Student in Artificial Intelligence
Supervised by Peidong Liu; Research: Spatial Intelligence, Embodied AI, Multimodal Learning

Wuhan University 2023 – 2026
Master in Pattern Recognition and Intelligent Systems
Research: Computer Vision, 3D Reconstruction, Multimodal Learning

Wuhan University 2019 – 2023
B.S. in Spatial Information and Digital Technology

RESEARCH PROFILE

I study how intelligent agents **build, predict, and act on representations of the physical world**. My work spans 3D scene reconstruction and generation, long-horizon video reasoning, and multimodal post-training, building toward spatially grounded, action-aware world models for embodied agents.

EXPERIENCE

ByteDance, TikTok Global Live LLM Strategy Department 2025 – 2026
Research Intern; Mentored by Xiangtai Li

- Co-developed **video MLLMs for temporal grounding**, with ownership of SFT/RL post-training pipelines, reward design, and evaluation.
- Built scalable data and annotation pipelines for high-quality, temporally grounded multimodal training data.
- Scaled distributed training and evaluation to **hundreds of GPUs**; analyzed training stability and model convergence.
- Designed agentic workflows leveraging LLMs to automate real-time interaction analysis and optimize content distribution within the TikTok Live ecosystem.

SELECTED PUBLICATIONS

SIU3R: Simultaneous Scene Understanding and 3D Reconstruction Beyond Feature Alignment 2025
NeurIPS 2025 Spotlight 160+ GitHub Stars 10+ Citations Project Page ArXiv Code
Qi Xu*, Dongxu Wei*, Lingzhe Zhao, Wenpu Li, Zhangchi Huang, Shunping Ji, Peidong Liu
Introduced a shared pixel-aligned 3D representation for simultaneous reconstruction and scene understanding from unposed images, providing a unified spatial state for geometry and semantics.
My Contributions: Project lead and primary code implementer; co-developed the core method and led benchmarking, ablation analysis, and paper writing.

Any 3D Scene is Worth 1K Tokens: 3D-Grounded Representation for Scene Generation at Scale 2026
arXiv 2026 Project Page ArXiv Code
Dongxu Wei*, **Qi Xu***, Zhiqi Li, Hangning Zhou, Cong Qiu, Hailong Qin, Mu Yang, Zhaopeng Cui, Peidong Liu
Introduced a compact 1K-token implicit 3D latent representation and 3D Diffusion Transformer for scalable scene generation.
My Contributions: Co-designed the core method; implemented the training code and led model training; built the large-scale training dataset and led ablations and evaluation.

Towards One-to-Many Temporal Grounding 2026
ICML 2026 Project Page ArXiv
Qi Xu*, Yue Tan*, Shihao Chen, Jiahao Meng, Anran Wang, Shunping Ji, Hao Fei, Xiangtai Li
Introduced the first systematic solution for OMTG, featuring a 56k-sample dataset and RL training with CoT-based rewards; outperformed Gemini 2.5 Pro by 15.85%.
My Contributions: Co-initiated the project and co-developed the core idea; owned data construction, SFT/RL pipelines, reward design, model training, benchmarking, and paper writing.

ADDITIONAL PUBLICATIONS

VideoZeroBench: Probing the Limits of Video MLLMs with Spatio-Temporal Evidence Verification
arXiv 2026 Project Page ArXiv Code
Jiahao Meng, Tan Yue, **Qi Xu**, Haochen Wang, Zhongwei Ren, Weisong Liu, Yuhao Wang, Renrui Zhang, Yunhai Tong, Haodong Duan — 2026
A hierarchical long-video QA benchmark that verifies model answers with temporal intervals and spatial boxes under a five-level evidence protocol.
My Contributions: Co-designed the five-level spatio-temporal evidence evaluation protocol; designed and implemented the annotation tool and guidelines; constructed approximately 200 complete QA-evidence samples.

Watch, Remember, Reason: Human-View Video Understanding with MLLMs
arXiv 2026 ArXiv GitHub
Jiahao Meng, Yue Tan, **Qi Xu**, Kuan Gao, Weisong Liu, Yanwei Li, Xiangtai Li, Lingdong Kong, Haochen Wang, Qianyu Zhou, Jiangning Zhang, Guangliang Cheng, Yunhai Tong, Lu Qi, Minghsuan Yang — 2026
Surveyed human-view video understanding with MLLMs through watching, remembering, and reasoning abilities for long, multimodal, and knowledge-intensive video scenarios.
My Contributions: organized literature for the Memory section, created figures and tables, and contributed to paper writing and revision.

HiCI: Hierarchical Construction–Integration for Long-Context Attention
ICML 2026 ArXiv Code
Xiangyu Zeng, **Qi Xu**, Yunke Wang, Chang Xu — 2026
Efficient long-context modeling using hierarchical attention, extending LLaMA-2 to 100K tokens with only 5.5% additional parameters.
My Contributions: participated in partial method design and paper revision.

E-MoFlow: Learning Egomotion and Optical Flow from Event Data via Implicit Regularization
NeurIPS 2025 Project Page ArXiv Code
Wenpu Li, Bangyan Liao, Yi Zhou, **Qi Xu**, Pian Wan, Peidong Liu — 2025
An unsupervised framework for joint ego-motion and optical flow estimation using implicit neural representations and differential geometric constraints.
My Contributions: participated in partial method design and paper revision.

RESEARCH MENTORSHIP

SupIR-GS: Thermal Infrared Super-Resolution Novel View Synthesis with Imaging-Calibrated 3DGS
ECCV 2026 Corresponding Author
Jin Liu, Haodong Li, Jiagang Chen, Dabin leng, Jiguang Li, Zhao Huang, Xiaoshuai Zhang, **Qi Xu**†, Zhiwen Zheng, Xingru Huang† — 2026
Reconstructs high-resolution 3D thermal scenes from low-resolution inputs via physics-informed degradation modeling.
Mentorship: Provided the core idea, guided method design and experiment planning, and led paper writing and revision.

Physically Grounded Dual-Opacity Gaussian Splatting for Joint RGB-TIR Reconstruction
ECCV 2026 Corresponding Author
Jin Liu, Dabin leng, Jiagang Chen, Haodong Li, Jiguang Li, Zhao Huang, Xiaoshuai Zhang, Zhiwen Zheng, Xingru Huang†, **Qi Xu**† — 2026
A unified RGB-TIR reconstruction framework resolving cross-spectral visibility conflicts through thermal field modeling.
Mentorship: Proposed the core idea, coordinated experiments and ablations, and led paper writing and revision.

SKILLS

Research: 3D reconstruction and generation, video reasoning, spatio-temporal grounding, multimodal SFT/RL post-training, benchmark construction, and test-time scaling.

Direction: Action-conditioned world models, physical reasoning, planning, and learning for embodied agents.

Engineering: Python, PyTorch, distributed training/evaluation, data construction pipelines, and large-scale experimental analysis; familiar with C++ and Linux development.

Writing & Leadership: Leading research projects from problem formulation to implementation, benchmarking, analysis, paper writing, and public release.